Tadiyos Begashaw Banti

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RESEARCH INTERESTS

I'm interested in exploring how AI can be used robustly and reliably in health, education, and sustainability. I am especially interested in making AI more consistent and resilient to changes in data and setting.

EDUCATION

Addis Ababa Science & Technology University || Addis Ababa, Ethiopia
Bachelor of Science in Software Engineering, **Cumulative GPA: 3.29/4.00(3.43 last two years) ||** *06/2022*
Capstone: *Malaria Detection System using CNNs on Light Blood Smear Images - ranked #2 Final Capstone Project, College of Engineering, 2022.*
Relevant Coursework: Probability and Statistics (A-), Research Methodology in Software Engineering (A-), Advanced Programming (A), Operating Systems (A-), Systems Programming (A-), Principles of Compiler Design (A), Formal Language and Automata Theory (A-), Computer Graphics (A), Computer System and Information System Security (A-), Data Communication and Computer Networks (A-).

A2SV (African to Silicon Valley) || Addis Ababa, Ethiopia
Data Structure and Algorithms Camp (Competitive Programming), (Link) || 02/2023 - 11/2023
Focused on competitive programming, algorithmic problem solving, and applied AI development.
Relevant Coursework: Sorting, Sliding Window, Prefix Sum, Recursion, Graph, Greedy, Dynamic Programming, Binary Search & Binary Trees

APPLIED PROJECTS

Malaria Detection System
Malaria detection system using thin blood smear images 09/2021 - 06/2022

- Took inspiration from an NIH research paper on training malaria parasite detection on thin blood smears to apply it to a local data source (Research Paper).
- Collected thin blood smear images from hospitals and clinics in Addis Ababa and preprocessed them (resizing, normalization, and cell segmentation) to extract **~6,000** balanced cell images.
- Trained a pretrained **ResNet-50** classifier with a patient-separated split, reaching **0.89** accuracy.
- Added a human review, where if the model was not at least **95%** confident a cell was uninfected, it was sent for human review.

Reducing Hallucination in a Production RAG system - AfroChat
A2SV - Project Lead & Software Engineer 09/2023 - 05/2024

- Built a RAG pipeline for internal documents using LangChain, PostgreSQL (pgvector), and an OpenAI embedding model (ada-002).
- Experimented with different guardrail methods like source verification, clarifying questions, and refusal with redirection, where refusal with redirection had the best outcome.
- Reduced hallucination rate by **40%,** measured through manually created Q&A and using GPT-4 as a judge for the GPT-3.5 outputs.
- Experimented with fine-tuning an open-source Llama model for the Amharic language by training it on different translated books from English to Amharic and vice versa.

Real-Time Network Malware Detection - INSA
INSA (Information Network Security Administration) - ML intern 08/2022 - 02/2023

- Worked on a hybrid malware detection system on internal networks by combining rule-based, role-based, and ML classifiers for unknown patterns.

WORK EXPERIENCE

Hydrus || Remote (Full-Time)
Full-Stack Software Engineer (Link) || 03/2025 - present

- Hydrus is a startup based in Silicon Valley, focusing on an intuitive sustainability management platform to solve sustainability data challenges.
- Built a multi-agent and tool-based LLM assistant for enterprise clients to query, analyze, and edit data through chat, which led to a significant drop in manual reporting effort.
- Cut LLM context per request from ~70 MB to under ~1 MB (**>98%**) by scoping context to the user's active page for better chat latency and smaller overhead in requests.
- Analyzed multi-year emission data to ensure calculation accuracy by mapping complex consumption surveys to specific emission factors. This led to resolving discrepancies of **1 - 5%** in reported emissions.
- Optimized searching and indexing of a global emission factor library of **800k+** emission factors to decrease the time in adding them to emission calculations.

beU (Delivery Platform in Ethiopia) || Remote (Part-time)
Senior Full-Stack Software Engineer (Link) || 07/2026 - present

- beU is a delivery platform incubated by Y-Combinator with **700k+** users and **40k+** monthly active users.
- Leading the end-to-end rewrite of the legacy system (PHP) to TypeScript and Flutter, from development to deployment of the system and system design decisions.

- Reduced AWS cost by **12%** by replacing full table scans of geospatial queries with spatial indexes and bounding-box pre-filtering, and deployed a self-hosted OSRM engine to cut the third-party API for routing from $1200/month to under $100/month (**>90%**).
- Replaced full manual review of refund requests with a rule-based system that checks request frequency, amount of the order, and other attributes. It enabled a **5%** decrease in fraudulent refunds and decreased the overload of requests for manual review by **70%.**
- (In Progress) Researching an image-based calorie and macronutrient estimation feature where users can get nutritional information from a picture of their meal.

A2SV - Africa To Silicon Valley || Addis Ababa, Ethiopia
Project Lead and Software Engineer II of AfroChat (Link) || 03/2023 - 02/2025

- Led a team of 12 engineers, a product manager & Lead UI/UX designer in developing web, mobile, and SaaS for different organizations and internal tools in A2SV.
- Mitigated different DDoS attacks by identifying and patching vulnerabilities and integrating different monitoring tools.
- Scaled to more than **85+** countries and **7000+** monthly active users through SEO optimization, Native language integration, and custom AI characters.
- Integrated LLM into A2SV's student tracking platform to provide personalized insights and progress reports for **~700** students, mentors, and admins.

Earlier Internships

Mn-Ale-Addis, Backend Developer Internship || 01/2020 - 03/2020

- Engineered scalable backend infrastructure using Django and MongoDB for an event-organizing platform, integrating web-based CRUD interfaces and secure user authentication.

Blue Labs, Full-Stack Developer Internship || 01/2019 - 03/2019

- Developed a web app that enables farmers to collect various data, such as humidity, water amount, etc., for in-house farms.

TEACHING AND MENTORING

DSA Course Instructor and Mentor *03/2024 - 02/2025*

- Taught advanced DSA topics (Graphs, Dynamic Programming, Binary Search) to multiple student cohorts.
- Mentored students and hackathon teams on software design, team management, and product development.
- Organized **A2SV Hackathons 2023 & 2024**, attracting **5,000+** participants from **1,000+** universities across 48 African countries.
- Conducted Generative AI training sessions for organizations exploring system integration with AI.

Student Mentor - Ethiopian Medical Advancement Initiative Student Association **AASTU**

- Mentorship of Medical and Engineering students in utilizing technology in the health sector.
- Contributed to developing a COVID-19 resource locator and distributing platform and showcasing how to utilize it in different Hospitals and Clinics.

TECHNICAL SKILLS

Languages: Proficient in Python, Java, SQL, C, C++, and JavaScript. Experience in PHP and Go
AI & ML: TensorFlow, PyTorch, Scikit-learn, LangChain, CNNs, RAG, LLM Evaluation, Transformers
Cloud & DevOps: Docker, AWS, GCP, Linux, Git, Kubernetes
Databases: SQL and NoSQL Databases (Postgres, MongoDB)
Tools & Frameworks: FastAPI, Flask, AutoGen, langchain, Git Workflow, Github Actions, React, Laravel, Flutter

AWARDS & HONORS

Top 3, A2SV Competitive Programming Camp 2023
Ranked #2 Final Capstone Project, College of Engineering, AASTU 2022
Top 20, Ethiopian Collegiate Programming Contest *2021*
Certificate of Appreciation, AASTU, Build internal scheduling system 2019

CERTIFICATIONS

Machine Learning by Stanford University (Coursera)